

CONCEPT 9.2

Glycolysis harvests chemical energy by oxidizing glucose to pyruvate

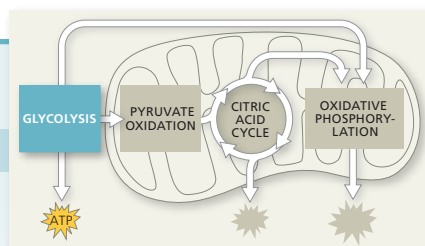
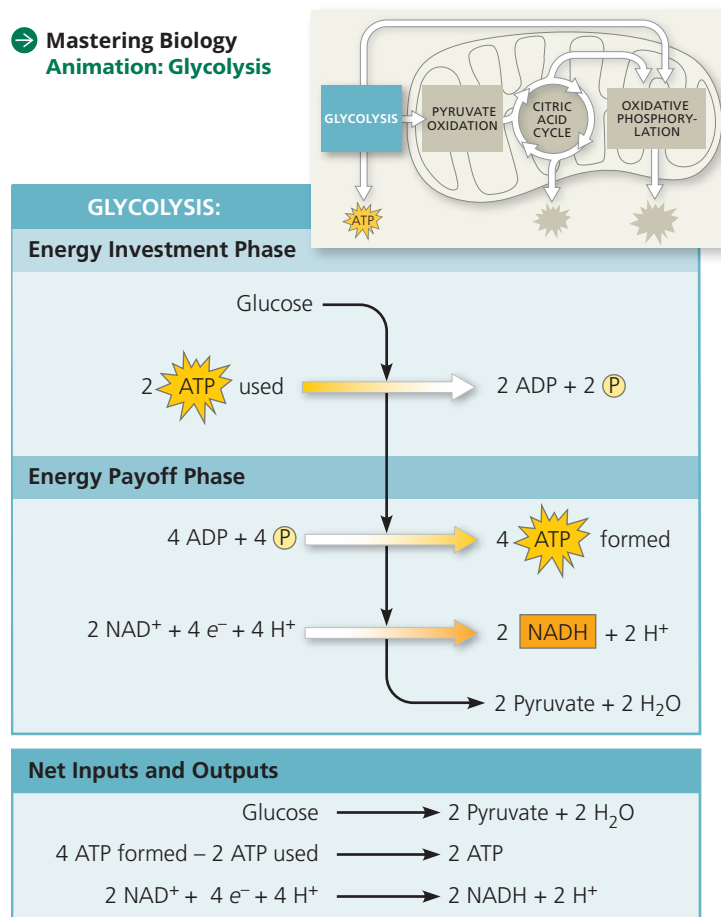
The word *glycolysis* means “sugar splitting,” and that is exactly what happens during this pathway. Glucose, a six-carbon sugar, is split into two three-carbon sugars. These smaller sugars are then oxidized and their remaining atoms rearranged to form two molecules of pyruvate. (Pyruvate is the ionized form of pyruvic acid.)

As summarized in **Figure 9.7**, glycolysis can be divided into two phases: the energy investment phase and the energy payoff phase. During the energy investment phase, the cell actually spends ATP. This investment is repaid with interest during the energy payoff phase, when ATP is produced by substrate-level phosphorylation and NAD^+ is reduced to NADH by electrons released from the oxidation of glucose. The net energy yield from glycolysis, per glucose molecule, is 2 ATP plus 2 NADH. The ten steps of the glycolytic pathway are shown in **Figure 9.8**.

All of the carbon originally present in glucose is accounted for in the two molecules of pyruvate; no carbon is released as CO_2 during glycolysis. Glycolysis occurs whether or not O_2 is present. However, if O_2 is present, the chemical energy stored in pyruvate and NADH can be extracted by pyruvate oxidation, the citric acid cycle, and oxidative phosphorylation.

▼ **Figure 9.7** The inputs and outputs of glycolysis.

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Animation: Glycolysis



▼ **Figure 9.8** The steps of glycolysis. Glycolysis, a source of ATP and NADH, takes place in the cytosol. Two of the enzymes (in steps 1 and 3) are shown in Figure 6.32b.

GLYCOLYSIS: Energy Investment Phase

WHAT IF? What would happen if you removed the dihydroxyacetone phosphate generated in step 4 as fast as it was produced?

